BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: PR IX (01) 21

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** Mathematics Paper–IX (01) (21)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. In ΔABC, if $\angle A \cong \angle B$, then bisector of ....divides the ΔABC into two congruent triangles.

(A) $\angle A$ (B) $\angle B$

(C) $\angle C$ (D) $A\bar{B}$

Q2. If sum of the measure of $\angle A$ and $\angle C$ of a parallelogram is 120°, then $m\angle B$ =

(A) 50° (B) 60°

(C) 65° (D) 120°

Q3. Perpendicular bisectors of a triangle are.....

(A) Congruent (B) Concurrent

(C) Parallel to each other (D) Perpendicular to each other

Q4. Which of the given represents the sides of a triangle?

(A) 3,4 and 5 (B) 3,4 and 7

(C) 3,4 and 8 (D) 3,4 and 1

Q5. Diagonal of a rectangle measures 6.5 cm. If its width is 2.5 cm. Its length is.....

(A) 4 cm (B) 9 cm

(C) 6 cm (D) 3 cm

Q6. HCF of $a^3 - b^3$ and $a^2 + ab + b^2$ is:

(A) $a - b$ (B) $a + b$

(C) $(a + b)(a - b)$ (D) $a^2 + ab + b^2$

Q7. Solve for x: $|x - 7| = 1$

(A) 8 (B) 6 and 8

(C) 8 and −8 (D) 6

Q8. The point (5, −2) located in quadrant is:

(A) I (B) II

(C) III (D) IV

Q9. $\sqrt{(2x+1)} - 5 = 4$ is a.....equation.

(A) Linear (B) Radical

(C) Cubic (D) Quadratic

Q10. If A(3,0) and B(0,3) are any two points in plane, then |AB| = ...

(A) 18 (B) $\sqrt{18}$

(C) $9\sqrt{2}$ (D) Zero

Q11. |4 −1; −9 2| = ...

(A) −1 (B) −17

(C) 1 (D) 17

Q12. $\sqrt[n]{(a^m)}$ = ...

(A) a^(n/m) (B) a^(mn)

(C) na^m (D) a^(m/n)

Q13. The characteristic of log 0.000059 is:

(A) −5 (B) 5

(C) −4 (D) 4

Q14. $(a + b)^2 - (a - b)^2$ = ...

(A) 4ab (B) $2(a^2 + b^2)$

(C) $(a + b)(a - b)$ (D) $a^2 - b^2$

Q15. Conjugate of $3 - \sqrt{5}$ is.....

(A) $-3 - \sqrt{5}$ (B) $-3 + \sqrt{5}$

(C) $3 + \sqrt{5}$ (D) $\sqrt{3} + \sqrt{5}$

Section-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 36

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

QII. Attempt any NINE parts. Each part carries FOUR marks.

1. Solve the system of linear equations $2x + 3y = -1$, $x - y = 2$ with the help of matrices.
2. Divide the first complex number by the second $Z_1 = 3i + 4$, $Z_2 = 1 - i$
3. Find the value of x, when $\log x = \bar{1}.8401$
4. If $x = 2 - \sqrt{3}$, find value of $x^4 + 1/x^4$
5. Write $2[x^2 + y^2 + z^2 - xy - yz - zx]$ as the sum of three squares.
6. Factorize $a^6 + b^6$
7. If H.C.F = $x + 5$ and L.C.M = $2x^3 + 11x^2 + 2x - 15$. Find polynomials of degree 2.
8. Find the square root of $4x^4 - 4x^3 + 13x^2 - 6x + 9$
9. Simplify $1/(a^2 + 7a + 10) + 1/(a^2 + 10a + 16)$
10. Find the solution set of $2\sqrt{2} = 17 + \sqrt{(40 - 3y)}$
11. The sum of three consecutive odd integers is 81. Find the numbers.
12. Draw the graph of the equation $y - 2x = 8$

Section-C

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 24

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII. Prove that points A(−2, −2), B(4, −2) and C(4, 6) are vertices of a right triangle.

QIV. If two angles of a triangle are congruent, then the sides opposite to those angles are congruent.

QV. Any point inside an angle, equidistant from its arms, is on its bisector.

QVI. Construct a ΔABC, when $m\overline{AB} = 4.5cm$, $m\overline{BC} = 3.1cm$, and $m\overline{CA} = 5.2cm$, draw their angle bisector and verify their concurrency.